

GLiTCH : GLiTCH induced Transitions for Secure Crypto-Hardware

C. Rohin Menon *, Jayanth Balasubramanian *, Akshay Kumar E, Annapurna Valiveti, Chester Rebeiro and Janakiraman Viraraghavan

Indian Institute of Technology Madras, India

Abstract—Conventionally, glitch reduction is well-studied in digital design to improve power, efficiency, and security. In contrast, this paper combines the addition and removal of glitches to minimize the power side-channel leakage. Glitch Manipulation is achieved through gate sizing-based arrival time control, which is cast as a Geometric Programming formulation. We develop a framework, GLiTCH, for glitch manipulation that is guided by functional and timing simulations. The framework is evaluated on popular cipher designs like AES, CLEFIA, and SM4. Our findings illustrate up to 52.82% improvement in the Guessing Entropy for a 38.74% area overhead on average across the evaluated ciphers.

Index Terms—Glitches, Side-channel attacks, geometric programming, gate-resizing.

I. INTRODUCTION

Glitches in digital circuits are unintended transitions at the output of logic gates that are caused by differential arrival times at the gates' inputs. Glitches present a significant challenge to the designer as they can contribute to an increase in the circuit's power consumption [1]. Besides these, glitches can significantly undermine the device's security. They are known to be the main cause of side-channel leakages in protected implementations of cryptographic ciphers. Due to these leakages, information about the secret key of ciphers can leak through side-channels such as the device's power consumption [2] and EM radiation [3]. Such attacks can compromise the secret key of ciphers like the AES within a few hours, even with strong countermeasures, like masking, in place [4].

Numerous approaches have been introduced to minimize power consumption resulting from glitch activity. These include gate sizing [5], [6] and selective gate freezing [7], as well as techniques that adjust input delays [8] and flip-flop insertions [9] to prevent or mitigate unnecessary transitions. Furthermore, several tools have been developed [10] to quantify and provide guidance on reducing glitch-induced power. In security-sensitive designs, alternative methods such as threshold cryptography [11] have been explored to address glitches. However, despite their efficacy, these techniques often face implementation challenges and generally introduce considerable area ($3\times$) and timing ($2\times$) overheads [12].

Contrary to popular belief that considers glitches as a curse that needs to be eliminated, we show in this paper that glitches

can, in fact, be harnessed to safeguard against side-channel attacks. We observe that glitches in a crypto-circuit are of two types: glitches that are correlated with the secret key and those that are not. Correlated glitches tend to increase side-channel leakage, while uncorrelated glitches act as noise, reducing the signal-to-noise ratio. Increasing uncorrelated glitches can, therefore, enhance side-channel security. On the other hand, by strategically altering the arrival times, correlated glitches can be reduced, enhancing the overall security of the design.

We propose a framework called GLiTCH, that, given the netlist of a cryptographic hardware design, uses gate sizing to reduce correlated glitches and increase uncorrelated glitches. GLiTCH works in two phases. The first phase quantifies the leakage of each gate through functional and timing simulations to identify critical gates in the given netlist. It then determines gate-sizing objectives in order to eliminate correlated glitches and increase uncorrelated glitches while adhering to user-specified timing constraints. The second phase performs glitch manipulation using a Geometric Programming (GP) formulation to control the arrival times of signals at the critical gates through gate-sizing [5]. The output of GLiTCH is a gate-sized netlist with reduced side-channel leakage.

The primary contributions of this work are summarized as follows:

- We present GLiTCH, a novel framework that leverages glitch manipulation to reduce side-channel leakage in RTL designs.
- Through GLiTCH, we demonstrate that in addition to reducing glitches, strategically enhancing glitches at specific gates in the design can aid in minimizing side-channel leakage.
- We validate the effectiveness of GLiTCH by evaluating it on widely used cryptographic ciphers, showcasing its ability to significantly reduce side-channel leakage.

The paper is organized as follows: Section II summarizes differential power analysis attacks and the necessary examples. The related work on glitch minimization and side-channel leakage optimization are summarized in Section III. We motivate the glitch manipulation strategy with the help of an example in Section IV. We present our framework in Section V followed by the results in Section VI. The final result concludes the paper.

* Both authors contributed equally. Corresponding authors contact: rohin@cse.iitm.ac.in, ee20b050@smail.iitm.ac.in.

TABLE I: The non-uniform distribution in Hamming weights leak information about the 2-bit k in AND gates, while the uniform distribution in EXOR gates do not leak information in EXOR gates.

m	pair of AND			
	k			
	00	01	10	11
00	00	00	00	00
01	00	01	00	01
10	00	00	10	10
11	00	01	10	11
$\sum$ HW	0	2	2	4

m	pair of EXOR			
	k			
	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00
$\sum$ HW	4	4	4	4

II. BACKGROUND

A. Differential Power Analysis

In a Differential Power Analysis (DPA) attack [2], an adversary correlates the power consumption traces collected from a crypto-device with a hypothetical power model to predict the cipher's secret key. Typically, Hamming Weight or Hamming Distance are used as the hypothetical power models. The success of a DPA attack depends on how well the secret key can be inferred from the device's power consumption. To understand this, consider a pair of EXOR and AND gates, with 2-bit inputs m and secret k . The truth tables for these gates are shown in Table I. Assume that m is uniform and that an adversary can observe the Hamming weight of each gate's output, then certain values of k can be inferred from the AND gate due to the non-uniform distribution of Hamming weights. On the other hand, the adversary cannot infer anything about k from the EXOR gate, due to the uniform distribution of Hamming weights. Thus, the signals in a design can be considered either to be *correlated* or *uncorrelated*. *Correlated* signals are the ones that can be used to infer information about k , while signals that cannot be used to glean information about k are called *uncorrelated* signals.

B. Glitches

If the inputs to a 2-bit NAND gate toggle from 0 to 1 and 1 to 0 respectively, then output remains at logic 1. However, in practice, the gate's inputs may arrive at different times, which might cause the output to briefly drop to logic 0 before recovering to the steady state logic 1 value, as seen in Figure 1. These spurious transitions are called glitches and may form if the arrival time imbalance at the gate's input exceeds the gate's inertial delay. For example, if the arrival for the inputs to the NAND gate are a_2 and a_1 respectively and its delay is d_1 , then glitches may form only if *i.e.* $|a_2 - a_1| \geq d_1$. As glitches propagate in the circuit, its effects may be noticed at other gates in the NAND gates fanout. Such glitches are usually considered a menace, considerably affecting the device's power consumption and side-channel leakage. However, in this paper, we investigate glitches to show that, if used carefully, glitches can enhance the device's security by reducing side-channel leakage.

III. RELATED WORK

As our work proposes a framework to minimize side-channel leakage using glitch manipulation, we recollect the related works in the domain of glitch manipulation and defense strategies from the side-channel community.

A. Glitch Minimization

Glitches can constitute up to 70% of the dynamic power [13] in a digital circuit, and hence, glitch reduction techniques have been explored in the prior art. Common techniques include gate freezing [7] and latch insertion [9], which require standard cell modification and ECO-like critical path modifications. Gate sizing [5], [6], [8] eliminates glitches by matching arrival times, mostly focusing on gate downsizing by localizing the timing slack. A more universal approach for gate sizing-based glitch reduction using Geometric Programming (GP) was proposed in [14], which allows both up-sizing and down-sizing while adhering to timing specifications. We adopt the GP formulation for reasons that will be explained in Section V.

B. Glitches and Side-channel Leakages

Cryptographic hardware is particularly susceptible to side-channel attacks, where adversaries exploit information leaked through power consumption to extract bits of a cipher's secret key. A widely adopted countermeasure is masking [15], which obfuscates sensitive variables in the implementation to conceal side-channel leakage. However, masking schemes are often vulnerable to glitches that correlate with the secret [4].

Threshold implementations mitigate leakage caused by glitches by decomposing the vulnerable function into multiple independent, non-interfering components, effectively nullifying the impact of glitches [11]. Despite their effectiveness, these techniques incur significant area and power overheads, and not all functions can be efficiently transformed into equivalent threshold implementations.

In this paper, we propose an alternative approach by examining the role of glitches in side-channel leakage. The subsequent section demonstrates the potential of controlled glitch manipulation as a novel and effective method for mitigating side-channel vulnerabilities.

IV. A MOTIVATING EXAMPLE

The success of a DPA attack depends on how well the adversary is able to correlate the observed signals to the secret key. We refer to the ratio of the number of correlated signals to the total number of observed signals, as the adversary's Distinguishing Advantage (DA). As an example, we use Figure 1 to motivate how glitches can affect the DA of an adversary at the output of the EXOR gate, *i.e.* at the wire n_6 .

We assume that every gate has unit delay and all inputs are initially at Logic '1'. In the circuit, consider m_1 and m_2 as two uniformly distributed values and k as a fixed secret value. Due to the varying logic depths (fan-in cones), each input may arrive at different times. This is captured by the arrival times a_1 , a_2 , a_3 , and a_4 that can take any value in $\{1\}$, $\{1, 3, 4, 5\}$, $\{2, 3, 4, 5\}$ and $\{2, 4, 5, 6\}$, respectively. The

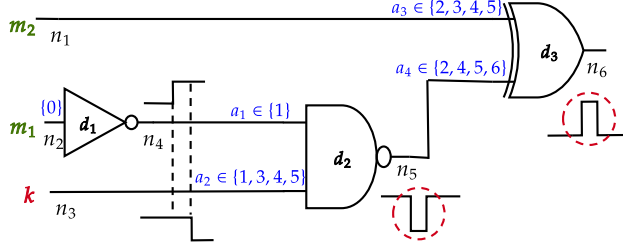


Fig. 1: Glitch generation with arrival times and signal transitions.

transitions observed at wire n_6 depend on the specific arrival times of signals at the wires n_1 and n_5 . These timing variations result in one of three transition patterns at n_6 :

- 1) **Case 1:** $a_3 = a_4$

$$\langle 1 \rightarrow \overline{m_1 \cdot k} \oplus m_2 \rangle$$

- 2) **Case 2:** $a_3 > a_4$

$$\langle 1 \rightarrow \overline{m_1 \cdot k} \oplus 1 \rightarrow \overline{m_1 \cdot k} \oplus m_2 \rangle$$

- 3) **Case 3:** $a_4 > a_3$

$$\langle 1 \rightarrow \overline{m_1} \oplus 1 \rightarrow \overline{m_1} \oplus m_2 \rightarrow \overline{m_1 \cdot k} \oplus m_2 \rangle$$

Among these transitions, Case 2 features a single correlated signal associated with $\overline{m_1 \cdot k} \oplus 1$, while the signal corresponding to $\overline{m_1 \cdot k} \oplus m_2$ is effectively masked by the uniform distribution of m_2 . In contrast, Cases 1 and 3 exhibit no correlated signals. Consequently, DA for Case 2 is $\frac{1}{2}$ (ignoring the initial signal), whereas for Cases 1 and 3, the DA is 0.

A. Effect of a glitch

Glitches can either aid or hinder an attacker, depending on their influence on the adversary's DA. We examine the dual impact of glitches, analyzing their effects on DA in both scenarios.

Glitch decreasing DA: Suppose that $a_3 > a_4$ and a glitch occurs on the wire n_1 . Then, the signal on the wire n_1 transitions as: $\langle 1 \rightarrow m_2 \rightarrow \overline{m_2} \rightarrow m_2 \rangle$. Due to this, the signal transitions at wire the n_6 are:

$$\langle 1 \rightarrow \overline{m_1 \cdot k} \oplus 1 \rightarrow \overline{m_1 \cdot k} \oplus m_2 \rightarrow \overline{m_1 \cdot k} \oplus \overline{m_2} \rightarrow \overline{m_1 \cdot k} \oplus m_2 \rangle$$

The glitch at n_1 has resulted in a reduced DA from $\frac{1}{2} = 0.5$ to $\frac{1}{4} = 0.25$. Hence, it becomes evident that a glitch on the preceding gate of the circuit could, in fact, introduce uncorrelated signals at subsequent nodes.

Glitch increasing DA: Suppose that $a_3 > a_4$ and a glitch occurs on the wire n_3 . Then, the signal on the wire n_3 transitions as $\langle 1 \rightarrow k \rightarrow \overline{k} \rightarrow k \rangle$. Then the transitions at wire n_6 are:

$$\langle 1 \rightarrow \overline{m_1 \cdot k} \oplus 1 \rightarrow \overline{m_1 \cdot k} \oplus 1 \rightarrow \overline{m_1 \cdot k} \oplus 1 \rightarrow \overline{m_1 \cdot k} \oplus m_2 \rangle$$

The corresponding DA has gone up from 0.5 to $\frac{3}{4} = 0.75$. We can observe that a glitch in the circuit may either increase or decrease the adversary's DA depending on the specific signal that glitches.

B. Gate sizing as a solution

A key insight from the preceding analysis is that the circuit does not leak if the computation of $\overline{m_1 \cdot k}$ occurs simultaneously with or after the arrival of m_2 , establishing the condition $a_4 \geq a_3$. *Gate sizing* offers an effective mechanism to control signal arrival times without modifying the logical implementation or introducing additional delay elements, such as buffers, into the circuit. In Figure 1, gate sizing is applied to the NAND gate to control the arrival of k , ensuring $a_2 \geq a_1$. This adjustment delays the computation of $\overline{m_1 \cdot k}$, ultimately ensuring $a_4 \geq a_3$ and minimizing the adversary's ability to distinguish key-related operations. Note that the condition $a_2 \geq a_1$ aligns with the occurrence of a glitch at the NAND gate.

Contemporary works [16], [17] have explored gate-sizing as a side-channel countermeasure. Karna [16], for instance, reconfigured gates in the design to reduce leakage occurring due to dynamic power consumption. On the other hand, [17] considered leakage due to static power consumption. However, neither study has examined the impact of glitches on side-channel leakage, which is the primary focus of this paper.

V. GLITCH: THE FRAMEWORK

This section presents the two phases of our framework, *leakage assessment* and *glitch manipulation formulation* while providing details about the steps involved in each phase.

The complete workflow of GLITCH is illustrated in Figure 2. Our analysis employs a leakage analysis tool to identify circuit leakage. Initially, functional simulation is performed on the gate-level netlist with zero gate delay, producing a leakage vector for each gate, denoted as L_{FN} . This zero-delay netlist is then analyzed using the leakage tool to quantify the circuit's *functional* leakage. Subsequently, the area-minimized (AM) version of the netlist is simulated with timing delays to capture practical gate behavior. The leakage analyzer estimates the leakage vector for this configuration as L_{AM} . Since the zero-delay simulation (FN) considers only functional transitions, the differences in leakage between FN and AM can be attributed exclusively to glitches introduced by timing delays. If the leakage in the AM circuit is lower than that in the FN circuit, it indicates the presence of randomizing glitches that act as noise, reducing leakage. To amplify this effect, we intentionally increase the disparity in arrival times at the corresponding gate to encourage the occurrence of such randomizing glitches. Conversely, if the AM leakage exceeds that of FN, it signifies the presence of correlated glitches, contributing to higher leakage. In this case, we minimize the difference in arrival times at the gate to suppress or eliminate these correlated glitches. This approach allows us to systematically identify critical gates whose glitch behavior significantly influences the overall leakage. We then

determine modification objectives for these critical gates based on the difference $L_{FN} - L_{AM}$. A non-negative value indicates the need to increase the disparity in signal arrival times at the gate, while a negative value necessitates bringing the arrival times closer together to achieve glitch reduction. This enhancement or reduction is achieved by gate sizing while ensuring compliance with the timing constraints of the AM circuit.

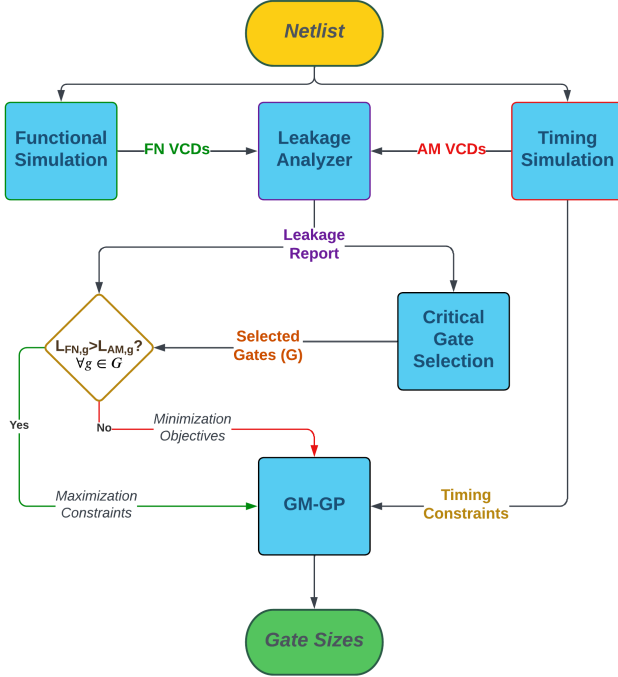


Fig. 2: GLITCH manipulation framework.

A. Leakage Analysis

To enhance side-channel resilience in designs, we propose a systematic approach for identifying and mitigating critical leakage points within the design. This process begins by evaluating the leakage characteristics of the design of both the functional and timing simulations. Each module in the design is analyzed individually, with power leakage quantified by examining signal transitions and their correlation with secret-dependent operations. The leakage analyzer operates in multiple phases: it first captures the signal transition traces (Value Change Dump files) from the functional and timing simulations and correlates these traces with the secret dependent operations to assess leakage. A leakage metric is computed based on this correlation, identifying the gates that contribute most significantly to leakage. The gates are then ranked according to their leakage impact, enabling the prioritization of countermeasures at the most critical points in the design's netlist.

B. Glitch Manipulation using Geometric Programming

We motivated the need for glitch manipulation using gate sizing in Section IV. In general, a GP optimization problem

is mathematically formulated as follows:

$$\begin{aligned} \min \quad & f_0(\mathbf{x}) \\ \text{subject to} \quad & f_i(\mathbf{x}) \leq 1, \quad i = 1, 2, \dots, m \\ & g_j(\mathbf{x}) = 1, \quad j = 1, 2, \dots, p \end{aligned} \quad (1)$$

where f_i are posynomial functions, and g_j are monomial functions [18]. Glitch manipulation (GM) can be categorized into two primary approaches: glitch addition and glitch removal. Within the selected critical gates, as outlined in the proposed framework, the choice to introduce additional glitches or to reduce existing ones is determined using the leakage analyzer. The GP formulation for both are discussed below:

1) Glitch Addition: We use the inertial delay model for the gates, according to which a gate does not respond to a signal if the next incoming signal (at the same input or another input) arrives within one inertial delay of the gate. Consequently, by ensuring that the difference in arrival times of signals at the gate inputs exceeds the inertial delay, the likelihood of generating additional glitches at the gate's output can be increased. For example, in Figure 1, the NAND2 gate requires $|a_1 - a_2| \geq d_{inert}$ for the gate to generate glitches, where d_{inert} is the inertial delay of the NAND2 gate. Without loss of generality, it is assumed $a_2 \geq a_1$. The equation can be rewritten as follows:

$$\frac{a_1 + d_{inert}}{a_2} \leq 1. \quad (2)$$

Equation (2) represents a valid posynomial expression that is suitable for inclusion in a geometric programming framework, as it adheres to the structural requirements of the formulation as outlined in 1.

2) Glitch Removal: To implement glitch removal, the inequality in equation (2) is to be reversed. However, this modification introduces a structural incompatibility with the formulation of the optimization problem as defined in equation (1). To ensure glitch removal, it is essential to minimize the difference in arrival times at the gate inputs. For the scenario depicted in Figure 1, this corresponds to reducing $(a_2 - a_1)$ so that the arrival times come close together. An objective compatible with the GP formulation for glitch removal [14] is as follows:

$$\min \left(\frac{a_2}{a_1} \right)^2, \quad \text{subject to constraint } a_1 \leq a_2.$$

Putting both the objective for the glitch removal, which can be extended for gates with more than two inputs [14] and the GP constraint for glitch addition into the formulation, we get the following set of equations:

The area minimization of the circuit, which is used as the baseline for timing simulations, can be realized by replacing the objective function with $\min(\sum_{i \in G} x_i)$ and retaining the AM-GP constraints from the above set of GM-GP formulation. The AM-GP solution provides the gate sizes that minimize area to meet a given timing specification.

Let,

- C : Set of all critical gates for glitch removal
- A_c : Set arrival times of critical gate (c) inputs for glitch removal
- A_m : Set arrival times at critical gate (c) inputs for glitch addition
- A_i : Set of all arrival times of the inputs to gate i
- G : Set of all gates
- O_i : Set of gate sizes of the fanout of gate i
- g_i : Logical effort of gate i
- p_i : Parasitic delay of gate i
- P_O : Set of all gates with a primary output
- x_{max} : Maximum size available in the library
- T_{spec} : Timing specification for the circuit

GM-GP:

$$\begin{aligned}
 & \min \quad \sum_{c \in C} \sum_{m \neq n} \left(\frac{a_n}{a_m} \right)^2 \\
 & \text{subject to} \quad a_m \leq a_n \\
 & \quad \quad \quad \frac{a_x + d_{inert}}{a_y} \leq 1 \\
 & \quad \quad \quad a_l + g_i \left(\frac{\sum_{x_j \in O_i} g_j x_j}{g_i x_i} \right) + p_i \leq a_i \\
 & \quad \quad \quad a_k \leq T_{spec} \\
 & \quad \quad \quad x_i \leq x_{max} \\
 & \quad \quad \quad x_i \geq 1 \\
 & \quad \quad \quad \forall a_l \in A_i, \forall i \in G, \forall k \in P_O \\
 & \quad \quad \quad \forall a_m, a_n \in A_c \\
 & \quad \quad \quad \forall a_x, a_y \in A_m \text{ \& } a_y \geq a_x
 \end{aligned}$$

AM-GP
constraints

VI. EXPERIMENTS AND DISCUSSION

We validate the GM-GP method on widely-used cipher designs like AES, CLEFIA, and SM4, with a particular focus on their S-Boxes, which are well-established as the primary source of leakage [19]. We conducted functional and timing simulations for each S-Box, selected critical gates in different configurations, determined their objectives, and applied the Glitch Manipulation methodology outlined in Section V.

A. Methodology

The total power and glitch count were analyzed using 1,000 uniformly sampled input vectors. These ciphers were synthesized using Cadence while employing the TSMC 28-nm standard cell library. The timing specification (T_{spec}) was carefully chosen between 1.2 and $1.4 \times T_{wall}$, where T_{wall} is the fastest achievable timing for the circuit [14]. Setting T_{spec} beyond this range would result in most gates becoming unit sizes. Simulations were performed on a 3.6 GHz Intel i7 system with 32 GB of memory running CentOS. The optimization framework was implemented in Python, utilizing the Gpkit library [20] alongside the Mosek solver [21].

The results presented in the paper validate the insights derived from the motivating example in Section IV, highlighting the critical role of glitches in leakage differences. In the motivating example, we demonstrated how variations in signal arrival times at a gate lead to correlated or uncorrelated transitions, directly influencing the leakage. The results obtained across FN, AM, and GM configurations provide empirical evidence supporting this explanation. By keeping the circuit structure and the number of gates constant across FN, AM, and GM, the results isolate glitches as the sole contributor to leakage variations. In addition, we assessed the Guessing Entropy (GE) [22] of the secret key for each scenario (FN/AM/GM), quantifying the number of attempts an attacker would need to correctly guess the key irrespective of the guessing strategy. This metric provides insights into the circuit's resilience to side-channel attacks in practical scenarios. Higher GE indicates stronger resistance, with an 8-bit S-Box limiting the range to integers between 1 and 256.

B. Results

The results of this evaluation are summarized in Table II, providing insights into the impact of gate modifications on the design across different configurations. The table details key parameters, including the number of modified gates, changes in glitch occurrences, the number of gates affected, the resulting Guessing Entropy (and therefore the DA), and the percentage increases in area and power consumption. Each entry for a chosen cipher corresponds to a varying choice of gates for glitch modification, allowing for a comparative analysis of their performance and security.

For instance, in the first configuration of the AES cipher, starting with a netlist of 195 gates, the design was altered by introducing 7 gate constraints for maximization and 5 gate objectives for minimization relative to the AM configuration. Maximization here refers to pulling apart signal arrival times to create intentional delays, while minimization aims to align arrival times more closely. These adjustments led to an increase of 8,446 glitches, with 115 gates experiencing a rise in glitches at their output and 69 gates showing a reduction. As a result, the GE rose to 232 from an initial 75, reflecting a 61.33% improvement. This increase in GE implies a lower DA, which correlates with reduced vulnerability to side-channel attacks. Notably, while this configuration incurs some area and power overheads (40.56% and 51.34%, respectively), it achieves the highest level of side-channel security among the evaluated setups, making it the most effective in mitigating leakage. Subsequent configurations, outlined in other rows of the table, present varying trade-offs in terms of the GE, DA, area, and power depending upon the decision made during the critical gate selection phase. Here, the first configuration, highlighted in green, demonstrates the best balance between security improvement and the incurred overhead, solidifying its role as the optimal choice for secure cryptographic hardware.

While the GE indicates the global effects of the modification, it is also interesting to study the impact of the changes on the leakage at a gate level. Figure 3 shows a comparison

TABLE II: Results of the proposed framework on various S-boxes indicating the cipher, Number of gates modified, Change in number of glitches, Guessing Entropy in DPA and % Area and Power overheads

Cipher	# Gates	# Gates Maximized	# Gates Minimized	Δ Glitches (GM - AM)	# Gates Glitch $\uparrow$	# Gates Glitch $\downarrow$	Guessing Entropy			% Area $\uparrow$	% Power $\uparrow$
							FN	AM	GM		
AES	195	7	5	8446 $\uparrow$	115	69	53	75	232	40.56	51.34
		0	5	45370 $\downarrow$	25	158			229	36.74	11.05
		1	0	26024 $\uparrow$	122	62			217	36.29	64.72
CLEFIA	156	4	0	18592 $\uparrow$	99	22	76	68	228	41.91	70.1
		4	0	18670 $\uparrow$	99	22			159	42.02	70.51
		5	0	15610 $\uparrow$	93	24			141	41.69	60.41
SM4	128	1	0	29914 $\downarrow$	29	95	214	105	209	34.62	7.54
		1	0	25682 $\downarrow$	42	83			205	35.76	29.85
		1	1	31250 $\downarrow$	36	89			154	39.09	27.89

of leakage scores for three different settings - FN, AM, and GM - across the range of signals present in the AES S-Box. The x-axis represents the signals with a select few marked for readability, while the y-axis shows the leakage score. The zoomed-in section of Figure 3 highlights the local and global effects of glitch modification on circuit behavior. Gate 1 and Gate 3 show a significant decrease in leakage scores, indicating a dominance of uncorrelated glitches. In contrast, Gate 2 exhibits a slight increase in the GM leakage score, suggesting a marginal rise in correlated glitches. This suggests that while glitch modification can lead to a reduction in overall glitches in some parts of the circuit, it can also result in an increase in correlated glitches in other parts. This can be attributed to the addition or removal of glitches in the fan-in or fan-out cones of the chosen gates due to changes in arrival times.

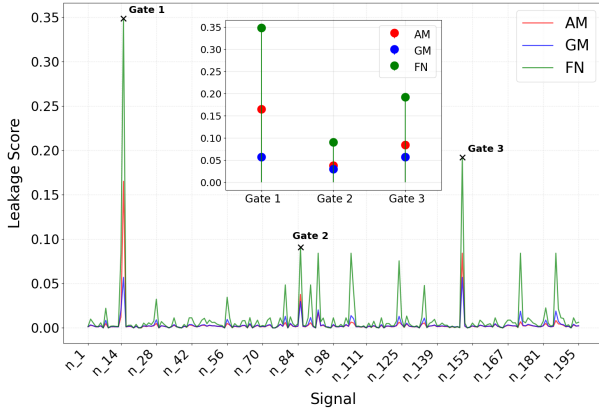


Fig. 3: AES Leakage Comparison

C. Discussion

The GM configuration, on average, incurs a 38.74% area overhead. Power, on the other hand, is dependent on the activity factor of each gate and its corresponding load capacitance. This is largely affected by the choice of critical gates, and hence, we observe that the power overhead ranges from 7 – 70% across the various cipher implementations. These overheads are accompanied by a notable reduction (52.82% on average) in the leakage and enhanced resilience, suggesting its potential suitability for secure cryptographic hardware. This result highlights the framework's ability to facilitate the design of hardware with stronger defenses against side-channel attacks. In this paper, we have summarized the

results for 8-bit S-Boxes. However, we have also evaluated ciphers with smaller S-Boxes like PRESENT and ASCON5. While we observed some reduction in the leakage, it came at the cost of large areas and power overheads. This can be attributed to the minimal number of re-sizable gates for the GM-GP formulation. For example, ASCON5 gave only a 9.38% improvement in the GE at the cost of 113.25% area and 104.27% power overheads.

Further, in this work, we consider relatively smaller circuits (128 – 195 gates) than a full cipher ($\sim 30k$ gates) to demonstrate a Proof-of-Concept. In general, the computational complexity of the AM-GP formulation scales as N^λ [23], where N is the number of gates in the circuit, and λ typically ranges between 1 and 2, depending on the circuit. Two kinds of constraints on a handful of gates are appended to the AM-GP formulation to incorporate the GM-GP formulation. As a result, the overall time complexity remains largely unaffected and should scale to a million gates as reported in [23].

VII. CONCLUSION AND FUTURE WORK

The GM configuration demonstrates a significant improvement in leakage reduction through targeted gate sizing. The results clearly demonstrate that the GE in the GM circuit is significantly higher compared to the AM circuit. Locally, the modifications directly impact the gates involved by altering their arrival times. Globally, these changes propagate throughout the entire fan-out network of the corresponding outputs in the circuit as the modified gates generate and transmit glitches. This can increase uncorrelated glitches across the circuit, which helps reduce side-channel leakage. The results show a 52.82% improvement in the GE with the 38.74% area overhead on average. Future directions involve understanding the global effects of the modifications on the leakages through the fan-in and fan-out cones. This insight would enhance the critical gate selection process.

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